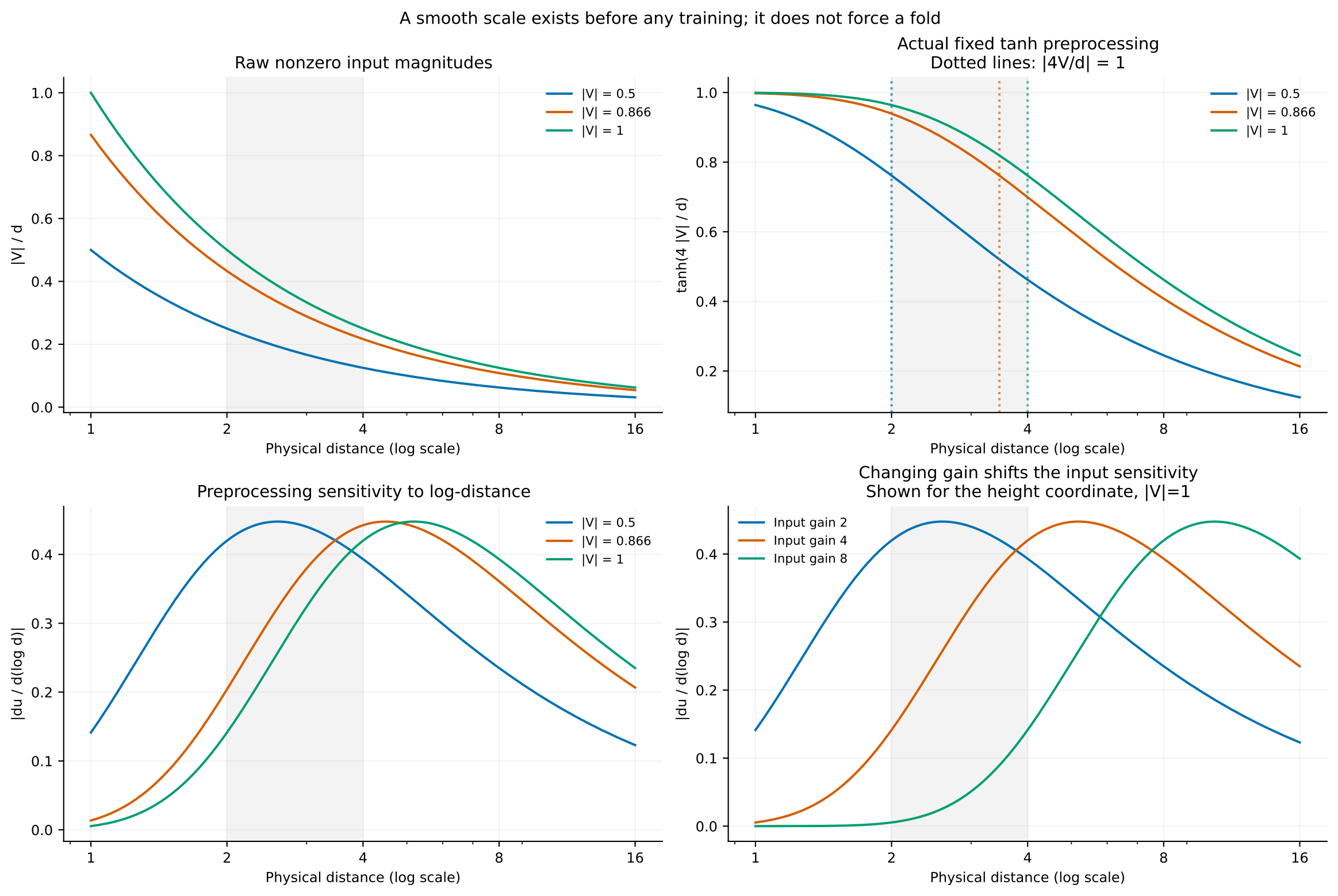
A smooth scale exists before any training; it does not force a fold
Raw nonzero input magnitudes
|V| / d
Physical distance (log scale)
|V| = 0.5
|V| = 0.866
|V| = 1
Actual fixed tanh preprocessing
Dotted lines: |4V/d| = 1
tanh(4 |V| / d)
Physical distance (log scale)
|V| = 0.5
|V| = 0.866
|V| = 1
Preprocessing sensitivity to log-distance
|du / d(log d)|
Physical distance (log scale)
|V| = 0.5
|V| = 0.866
|V| = 1
Changing gain shifts the input sensitivity
Shown for the height coordinate, |V|=1
|du / d(log d)|
Physical distance (log scale)
Input gain 2
Input gain 4
Input gain 8